PRODUCT USER MANUAL

For Baltic Sea Ice Multi-Year Observation Time Series

SEAICE_BAL_PHY_L4_MY_011_019

Issue: 1.1

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RECORD TABLE

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
MDS	Marine Data Store
RMS	Root mean square
PC	Production Centre
PU	Production Unit
SIC	Sea ice concentration
SIT	Sea ice thickness
IC	Ice chart
ITC	Ice thickness chart
ASIC	Automated SIC
ASIT	Automated SIT
SIE	Sea ice extent

SID	Sea ice drift
IDC	Ice drift chart
CSK	Cosmo-SkyMED, Itlaian X-band SAR (constellation)
TSX	TerraSAR-X, German X band SAR
PAZ	Spanish X-band SAR
RCM	Radarsat constellation mission, Canadian C-band SAR constellation
RS-2	Radarsat-2, Canadian C-band SAR
S-1	Sentinel-1. C-band SAR constellation of ESA
FMI	Finnish Meteorological Institute
SMHI	Swedish Meteorological and Hydrological Institute

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This user manual describes the regional Baltic sea ice product provided by the Copernicus Marine Sea Ice Thematic Assembly Centre (SITAC). The product SEAICE_BAL_PHY_L4_MY_011_019 multi-year datasets described in this document are based on the daily Baltic Sea ice charts. These product datasets are produced by FMI and disseminated through the central Copernicus Marine dissemination unit. They provide high resolution multi-year sea sea ice information over the Baltic Sea region. The data are freely available under the Copernicus Marine data license.

This product includes two datasets.

SEAICE_BAL_PHY_L4_MY_011_019 datasets:

- cmems_obs-si_bal_seaice-conc_my_1km: Gridded sea ice concentration, sea ice extent and classification based on the digitized Baltic ice charts produced by the FMI/SMHI ice analysts. It is produced daily in the afternoon, describing the ice situation daily at 12 UTC. The nominal resolution is 1km. The temporal coverage is from the beginning of the season 1980-1981 (Nov 1980) until today.
- cmems_obs-si_bal_phy-sit_my_l4-1km_P1D-m : Gridded sea ice thickness (in cm) from the beginning of the season 1980-1981 until today. The structure of the data is similar to the structure of FMI-BAL-SEAICE_THICK-L4-NRT-OBS product. The resolution and temporal coverage are similar as for the SIC dataset.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

The area covered by the products is the Baltic Sea. Both datasets cover the whole area. The Baltic Sea area is shown in Figure 1.

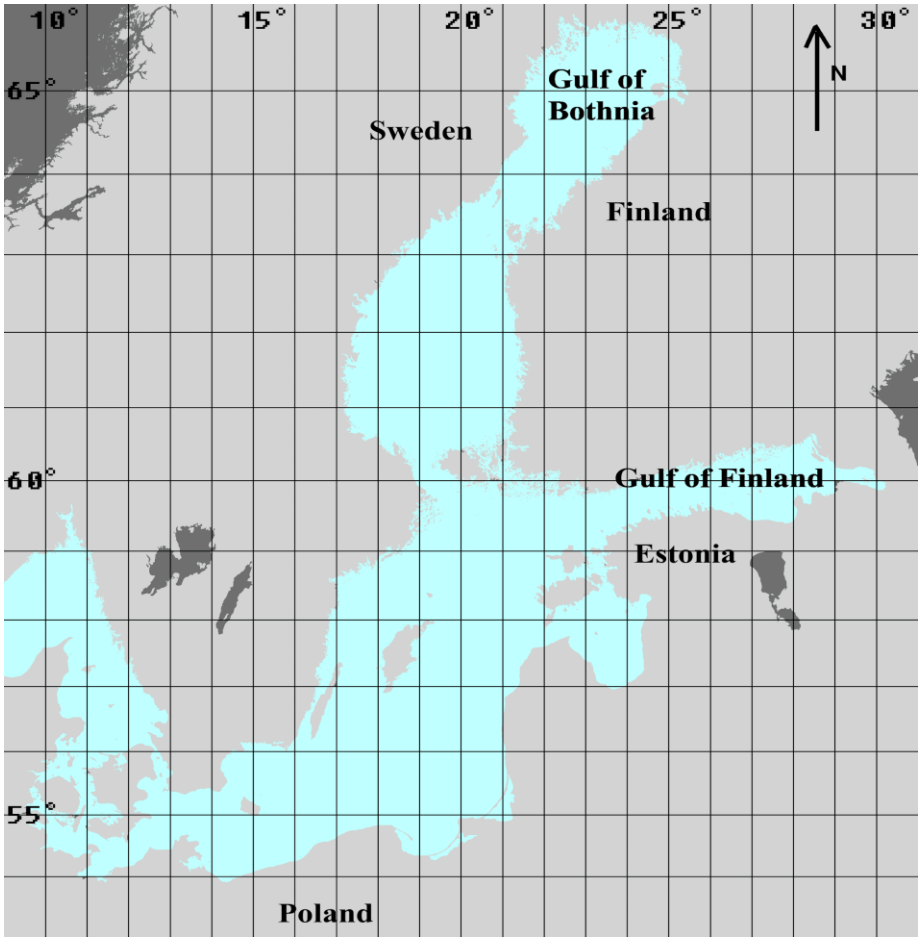

Figure 1. The coverage of the data: Baltic Sea area is indicated by the light blue color.

Product	Baltic Sea Ice SEAICE_BAL_PHY_L4_MY_011_019
Geographical coverage	Baltic Sea : 9°E - 31°E; 53°N – 66°N
Variables	• Sea Ice Concentration • Sea Ice Thickness • Sea Ice Classification
Product Type	Multi-Year - Observations
Available time series	Nov 1980-today
Temporal resolution	Daily

Update Frequency	Annually after ice season (early June)
Target delivery time	15 June
Resolution	1 km
Number of vertical levels	1
Format	NetCDF4

b) Production System Description

For the Baltic Sea- The operational sea ice service at FMI provides ice parameters over the Baltic Sea. The product is based on ice charts produced on a daily basis during the Baltic Sea ice season and show the ice concentration in a 1 km grid.

The operational sea ice service at FMI produces sea ice parameters based on a visual interpretation of satellite data and ground truth. The data are available since 3 November 1980 (season 1980-1981). The products are weekly until the season 2006-2007. From 2006-2007 the products are daily. The ice charts are based on satellite data and coastal, airborne and ship observations. In the beginning of the production period (since autumn 1980) the ice charts were based on TIROS-N/NOAA VIS/IR satellite imagery (Seinä et al. ,1997) , on airborne, coastal and ship (Finnish icebreakers) monitoring. In the season 1991/1992 SAR satellite data first from ERS-1 (launched July 1991), then ERS-2 from the season 1995/1995 (launched April 1995), Radarsat-1 from the season 1996-1997 (launched Nov 1995), ENVISAT ASAR from the season 2002-2003 (launched March 2002), Radarsat-2 from the season 2008-2009 (launched Dec 2007) and Sentinel-1 data from the season 2014-2015 have been used in the ice charting. Since late 1990's SAR instruments have been the major data source for the Baltic ice charting. Additionally coastal and ship observations have been utilized. Airborne observation was stopped in late 1990's and replaced by SAR data.

Sea ice concentration is given in %, sea ice extent in km2 and sea ice classes are as follows:

1 -> no ice or very open ice,
2-> open ice cover (SIC from 4/10 to 7/10)
3 -> close, very close and fast ice

Sea ice thickness is given in cm at each grid point. The product (both datasets) is weekly until the season 2006-2007 and daily starting from 2006-2007 (during the Baltic Sea ice season).

The product datasets have a fixed grid of 1223 (x) by 1445 (y) grid points, int latlong coordinate system. The upper left corner is (x,y)=(9.0,66) degrees, and the resolution is (dx,dy)=(0.018,0.009) degrees=(1km,1km). The reference ellipsoid of the coordinate system is WGS84.

c) Dataset details

Sea ice concentration, sea ice extent, sea ice classification:

Product	SEAICE_BAL_PHY_L4_MY_011_019
Data set	cmems_obs-si_bal_seaice-conc_my_1km
Geographical coverage	Baltic Sea 9°E -> 31° E ; 53°'N -> 66°N
Variables	Sea Ice concentration (%), sea ice extent (km2 sea ice classification (class)
Analysis	Yes
Forecast	No

Available time series	From Nov 1980 –> May 2024, updated annuall 15 June
Temporal resolution	Week, day since Dec 2006
Target delivery time	Aannually, after season, 15 June
Delivery mechanism	Subsetter, Copernicus Marine FTP
Resolution	1km
Number of vertical levels	1
Format	Netcdf CF1.4

Sea Ice thickness:

Product	SEAICE_BAL_PHY_L4_MY_011_019
Data set	cmems_obs-si_bal_phy-sit_my_l4-1km_P1D-m
Geographical coverage	Baltic Sea 9°E □ 31° E ; 53°'N □ 66°N
Variables	Sea Ice thickness (cm)
Analysis	Yes
Forecast	No
Available time series	From Nov 1980–> May 2024, updated annually June
Temporal resolution	Week, day (from Dec 2006)
Target delivery time	Annually after season, 15 June
Delivery mechanism	Subsetter, Copernicus Marine FTP
Resolution	1km
Number of vertical levels	1
Format	Netcdf CF1.4

SEAICE_BAL_PHY_L4_MY_011_019	
DATASET	**Variable's name in the NetCDF file and Unit** **Long_name** **CF Standard_name**
cmems_obs-si_bal_seaice-conc_my_1km	ice_concentration [%] FMI ice chart sea ice concentration sea_ice_area_fraction
	sea_ice_entent [km²] Sea ice extent

	sea_ice_extent
	sea_ice_classification [class] sea_ice_classification
cmems_obs-si_bal_phy-sit_my_l4-1km_P1D-m	ce_thickness [cm] FMI ice chart sea ice thickness sea_ice_thickness
	sea_ice_thicknsess range [cm] Sea ice thickness-range sea_ice_thickness_range

d) Additional information on some parameters

N/A

e) Processing information

Product disseminating schedule :

Type of product	Production time	Product temporal coverage (hours)	Product disseminating scheduled time	Dissemination day
SIC/SIE/sea ice classification observation grid time series	Annually in June	Whole season (dec-May)	Annually June 15	June 15
SIT observation grid time series	Annually in June	Whole season (dec-May)	Annually June 15	June 15

f) Catalogue updating

Daily, after production, during the ice season.

g) Time averaging

N/A

h) Product use and impact

The major use for the data are winter navigation and other offshore and coastal operations, e.g. off-shore and coastal construction. Can also be used e.g. in numerical weather and ice model data assimilation and evaluation.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

An exhaustive description of the files nomenclature when downloaded through the web portal is available here : Files Nomenclature and Format

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

5) FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
cmems_obs-si_bal_seaice-conc_my_1km	ice_conc_baltic_YYYYMMDD1400_MY.cdl	0.011
cmems_obs-si_bal_phy-sit_my_l4-1km_P1D-m	ice_thic_baltic_YYYYMMDD1400_MY.cdl	0.009

6) STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

7) REFERENCES

Quality Information Document
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SEAICE-QUID-011-004-011-019.pdf

A. Seinä, K. Palosuo, H. Grönvall, Merentutkimuslaitoksen jääpalvelu 1919-1994 (Ice Service of FIMR 1919-1994), published in Finnish Institute of Marine Research (FIMR) report series «Meri», in Finnish, 1997.

8) ANNEX

a) Structure and semantic of NetCDF maps files

These are examples of the file headers derived from the operational netcdf files:

```
011_019 MY SIC/SIE/classification
-----------------------------------------------------------------------
netcdf ice_conc_baltic_202405301400 {
dimensions:
      time = 1 ;
      lat = 1445 ;
      lon = 1223 ;
variables:
      int time(time) ;
            time:long_name = "reference time of the product" ;
            time:standard_name = "time" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:calendar = "standard" ;
            time:axis = "T" ;
      float sea_ice_extent(time) ;
            sea_ice_extent:long_name = "sea ice extent" ;
            sea_ice_extent:standard_name = "sea_ice_extent" ;
            sea_ice_extent:units = "1e6km2" ;
      float lat(lat) ;
            lat:long_name = "latitude" ;
            lat:units = "degrees_north" ;
            lat:standard_name = "latitude" ;
            lat:axis = "Y" ;
      float lon(lon) ;
            lon:long_name = "longitude" ;
            lon:units = "degrees_east" ;
            lon:standard_name = "longitude" ;
            lon:axis = "X" ;
      int crs ;
            crs:grid_mapping_name = "latitude_longitude" ;
            crs:false_easting = 0.f ;
            crs:false_northing = 0.f ;
            crs:proj4_string = "proj=latlong datum=WGS84" ;
      short ice_concentration(time, lat, lon) ;
            ice_concentration:long_name = "FMI ice chart sea ice
concentration" ;
            ice_concentration:standard_name = "sea_ice_area_fraction"
;
            ice_concentration:units = "%" ;
            ice_concentration:coordinates = "lon lat" ;
            ice_concentration:grid_mapping = "crs" ;
            ice_concentration:source = "fmi.fi" ;
            ice_concentration:_FillValue = -99s ;
      short concentration_range(time, lat, lon) ;
            concentration_range:long_name = "ice concentration range"
;
            concentration_range:units = "%" ;
```

```
            concentration_range:coordinates = "lon lat" ;
            concentration_range:grid_mapping = "crs" ;
            concentration_range:source = "fmi.fi" ;
            concentration_range:_FillValue = -99s ;
      byte sea_ice_classification(time, lat, lon) ;
            sea_ice_classification:long_name = "FMI ice chart sea ice
classification" ;
            sea_ice_classification:standard_name =
"sea_ice_classification" ;
            sea_ice_classification:_FillValue = 0b ;
            sea_ice_classification:valid_min = 1b ;
            sea_ice_classification:valid_max = 3b ;
            sea_ice_classification:flag_values = 1b, 2b, 3b ;
            sea_ice_classification:flag_meanings = "openwater openice
closeice" ;
            sea_ice_classification:flag_descriptions = "\n",
                  " 1 -> no ice or very open ice\n",
                  " 2 -> open ice cover (4 to 7 tenths) ice\n",
                  " 3 -> close, very close and fast ice\n",
                  "" ;

// global attributes:
            :title = "Baltic Ice Concentration and classification, L4,
1km daily" ;
            :Conventions = "CF-1.4" ;
            :product_name = "Baltic Sea Ice Concentration" ;
            :product_version = "1.0" ;
            :abstract = "Baltic Sea Ice Concentration (IC) and
classification based on the FMI ice chart." ;
            :area = "Baltic Sea" ;
            :institution = "Finnish Meteorological Institute (FMI)" ;
            :contact = "juha.karvonen@fmi.fi" ;
            :operational_status = "operational" ;
            :references = "" ;
            :history = "" ;
            :netcdf_version_id = "3.6.3" ;
            :southernmost_latitude = 53.2f ;
            :northernmost_latitude = 66.2f ;
            :westernmost_longitude = 9.f ;
            :easternmost_longitude = 31.f ;
            :start_date = "2024-05-30T14:00:00Z" ;
            :stop_date = "2024-05-30T14:00:00Z" ;
            :creation_date = "2024-06-06T15:36:15Z" ;
            :produced_date = "2024-06-06T15:36:15Z" ;
            :valid_date = "2024-05-30T14:00:00Z" ;
}

----------------------------------------------------------------------
011_019 MY SIT
----------------------------------------------------------------------
netcdf ice_thic_baltic_202405301400 {
dimensions:
      time = 1 ;
      lat = 1445 ;
                                    lon = 1223 ;
```

```
variables:
      int time(time) ;
            time:long_name = "reference time of the product" ;
            time:standard_name = "time" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:calendar = "standard" ;
            time:axis = "T" ;
      float lat(lat) ;
            lat:long_name = "latitude" ;
            lat:units = "degrees_north" ;
            lat:standard_name = "latitude" ;
            lat:axis = "Y" ;
      float lon(lon) ;
            lon:long_name = "longitude" ;
            lon:units = "degrees_east" ;
            lon:standard_name = "longitude" ;
            lon:axis = "X" ;
      int crs ;
            crs:grid_mapping_name = "latitude_longitude" ;
            crs:false_easting = 0.f ;
            crs:false_northing = 0.f ;
            crs:proj4_string = "proj=latlong datum=WGS84" ;
      short ice_thickness(time, lat, lon) ;
            ice_thickness:long_name = "FMI ice chart sea ice thickness
in cm" ;
            ice_thickness:standard_name = "sea_ice_thickness" ;
            ice_thickness:units = "cm" ;
            ice_thickness:coordinates = "lon lat" ;
            ice_thickness:grid_mapping = "crs" ;
            ice_thickness:source = "fmi.fi" ;
            ice_thickness:_FillValue = -99s ;
      short thickness_range(time, lat, lon) ;
            thickness_range:long_name = "ice thickness range in cm" ;
            thickness_range:units = "cm" ;
            thickness_range:coordinates = "lon lat" ;
            thickness_range:grid_mapping = "crs" ;
            thickness_range:source = "fmi.fi" ;
            thickness_range:_FillValue = -99s ;
      short product_quality(time, lat, lon) ;
            product_quality:long_name = "product quality" ;
            product_quality:units = "%" ;
            product_quality:coordinates = "lon lat" ;
            product_quality:grid_mapping = "crs" ;
            product_quality:source = "fmi.fi" ;
            product_quality:_FillValue = -99s ;

// global attributes:
            :title = "Baltic Ice Thickness, L4, 1km daily" ;
            :Conventions = "CF-1.4" ;
            :product_name = "Baltic Sea Ice Thickness" ;
            :product_version = "1.0" ;
            :abstract = "Baltic Sea Ice Thickness (SIT) based on the
FMI ice chart." ;
            :area = "Baltic Sea" ;
```

```
            :institution = "Finnish Meteorological Institute (FMI)" ;
            :contact = "juha.karvonen@fmi.fi" ;
            :operational_status = "operational" ;
            :references = "" ;
            :history = "" ;
            :netcdf_version_id = "3.6.3" ;
            :southernmost_latitude = 53.2f ;
            :northernmost_latitude = 66.2f ;
            :westernmost_longitude = 9.f ;
            :easternmost_longitude = 31.f ;
            :start_date = "2024-05-30T14:00:00Z" ;
            :stop_date = "2024-05-30T14:00:00Z" ;
            :creation_date = "2024-06-06T15:14:02Z" ;
            :produced_date = "2024-06-06T15:14:02Z" ;
            :valid_date = "2024-05-30T14:00:00Z" ;
}
```